

Stability of Haptic Systems with Fractional Order Controllers

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Abstract—Fractional order calculus is a generalization of the familiar integer order calculus in that, it allows for differentiation/integration with orders of any real number. The use of fractional order calculus in systems and control applications provides the user an extra design variable, the order of differointegration, which can be tuned to improve the desired behavior of the overall system. We propose utilization of fractional order models/controllers in haptic systems and study the effect of fractional differentiation order on the stability robustness of the overall sampled-data system. Our results demonstrate that fractional calculus generalization has a significant impact on both the shape and area of stability region of a haptic system and inclusion of fractional order impedances may improve the stability robustness of haptic rendering. Our results also include experimental verification of the stability regions predicted by the theoretical analysis.

I. INTRODUCTION

The main goal in haptic rendering is to enable a human operator to safely and transparently manipulate a virtual environment, through interactions with a robotic interface. During haptic rendering, stability robustness is crucial for ensuring the safety of the interaction, while transparency is required for the interaction to feel natural. However, since stability robustness and transparency are competing design objectives, achieving an optimal trade-off between these two criteria is one of the major research themes in the field of haptics.

This paper investigates the stability properties of haptic systems with fractional order controllers/models in their virtual environment. In particular, we analyze the stability regions of haptic systems for different differentiation orders and investigate the robustness of fractional order controllers to the changes in the human operator model.

Fractional order calculus is a generalization of the familiar integer order calculus in that, it allows for differentiation/integration, called differointegration, with orders of *any real number*. The use of fractional order calculus in systems and control applications provides the user an extra design variable, the order of differointegration, which can be tuned to improve the desired behavior of the overall system. Fractional order models are commonly used for modeling viscoelastic materials, while fractional order controllers are widely employed for motion control, thanks to their favorable robustness properties.

Intuitively, a fractional order derivative behaves as an interpolation between the neighboring integer order derivatives,

due the continuous behavior of the differointegration operator with respect to its order. For instance, considering position signal as the input, continually varying the order of differentiation order from 1 to 0 acts as changing the properties of a linear mechanical element from a pure dissipation element towards a pure potential energy storage (stiffness) element. Likewise, tuning the differentiation order from 1 to 2 acts as continually transforming from a pure dissipation element towards a pure kinetic energy storage (inertia) element. Note that dissipation exists for all differentiation orders in the open set $(0, 2)$, while pure energy storage takes place only for the integer orders of 0 and 2.

The rest of the paper is organized as follows. In Section II a literature review on the stability analysis of haptic systems is provided. Section III summarizes some of the fundamental properties of fractional order calculus and control systems. Section IV introduces the haptic system architecture used for the stability analysis, while Section V provides the stability analysis of fractional order haptic systems and validates these results through physical experiments. Finally, Section VI concludes the paper.

II. RELATED WORK

A haptic system is desired to stay stable at all times, for any human operator, and under any operation/grip conditions. The presence of the human operator in the loop significantly complicates the coupled stability analysis and controller design of haptic systems. First and foremost challenge is to find a simple and reliable model for the human operator, since without a model of the human operator, determining the coupled stability of the haptic system is not a trivial task. Furthermore, the sampled-data nature of the haptic systems introduces an additional challenge to the analysis.

The coupled stability analysis of haptic systems can be loosely categorized into two different approaches. The first approach focuses on the haptic system alone and aims at robust stability of the haptic system for a certain, but a wide, range of human operator models. On the other hand, the second approach assumes a model for the human operator and checks for the overall stability of the system based on this model. Former is considered as a more reliable analysis method resulting in conservative results, while the latter method is less conservative and especially useful to establish an understanding of the underlying relations among parameters of a haptic system on the overall stability of the system.

Robust stability based approaches generally rely on methods that do not require a specific model for the human operator; stability is considered for all possible human models

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within a certain class; therefore, the results obtained are generally more robust, but also more conservative, compared to analysis based on human models. The pioneering method in this branch is based on passivity analysis. The passivity based methods assume that the human operator behaves as a passive element. This widely accepted assumption is based on [1], where it has been shown that within the frequency range required for the haptic applications, humans generally act as passive network elements. In the teleoperation field, passivity analysis is first applied by Anderson and Spong [2] on continuous time models. Later, in his seminal work, Colgate introduced the passivity theorem for sampled-data system and applied it to haptics [3]. Without the need for a human model, the overall haptic system is rendered passive, so that any possible instability that might occur are avoided. Colgate's theorem had a profound impact in the field, since it handles the haptic system as a sampled-data system. Later, the passivity approach is generalized for interactions with unknown passive virtual environments through the idea of virtual coupling [4]. Virtual coupling acts as a buffer between the virtual environment and the robot; moreover, the parameters of the coupler are selected such that the robot-coupler 2-port network is passive. The virtual coupler idea is further extended in [5] with unconditional stability theorem for 2-port networks and to include admittance type devices.

Coupled stability analysis based on human models have also been adapted since the early studies in haptics. In these works, researchers have assumed simple, typically second order linear time invariant (LTI), models to represent the human operator in the loop. The pioneering work on the haptic system stability has been presented by Minsky *et al.* [6], where the human operator is approximated by a second order LTI model, while the discrete elements, sampler and hold, are approximated with continuous time models. Nyquist stability criterion is used to determine the stability of the overall system. In [7], Gillespie *et al.* have adopted a similar approach and shown that, the switching nature of the virtual wall and its discrete time implementation cause energy leaks and this leakage may result in instability. Stability analysis based on human models have also been employed to study the effects of various design variables of haptic systems on stability robustness. In particular, in [8], Routh-Hurwitz criteria is used to characterize the uncoupled stability of the haptic system, where no human operator is considered. Further analysis is conducted in [9] to investigate the effects of physical damping, time delay, human operator on stability properties. A Lyapunov based approach is introduced in [10] to determine the stability of the haptic system and to study the effects of discretization, quantization, time delay and Coulomb friction on the stability robustness.

Alternative approaches to study the coupled stability of haptic interactions exist and most notable of these studies include the time domain passivity approach due to Hannaford and Ryu [11]–[13] and the bounded impedance analysis by Haddadi and Hastrudi-Zaad [14].

III. FRACTIONAL ORDER CONTROL

Fractional order calculus has its roots in the late 17th century, that is, it is a peer of the celebrated calculus with integer order derivatives and integrals. Even though the idea of fractional order calculus can be found in the letters of L'Hopital, Leibniz and Bernoulli, the true development of fractional order calculus falls into the 18th and 19th centuries.

A fractional differointegral, fills the gap between the consecutive integer order derivatives/integrals. In other words, in terms of the differentiation/integration, derivative/integral operator is not discrete; instead, there is a smooth transition between successive orders of differentiation/integration. The idea of differentiating/integrating a function with an arbitrary order is, at first sight, counterintuitive, but it is a phenomenon that has been observed in nature. For instance, modeling viscoelastic materials using fractional order calculus is known to yield better results than modeling these materials with integer order calculus. In [15], it has been shown that using fractional order models can significantly reduce the degree of the model. In [16], a fractional order model is proposed for red blood cells.

Fractional order control has also found wide applications in the field of robotics and controls. A fractional order controller, called CRONE, can be designed to exhibit iso-damping behavior, even if the parameters of the system are changed significantly [17]. This control method has been successfully implemented in industrial applications, such as in car suspensions. A fractional order counterpart of the reputable PID controller is also commonly employed in motion control applications [18]. Tilted integral derivative (TID) is another commonly used fractional calculus based controller [19]. A quantitative comparison of these controllers with respect to their integer order counterparts can be found in [20]. Fractional order controllers are mostly used robust motion control [21]; however, they have been also applied to position-force hybrid control in [22]. There exists many notable books covering fractional order control [23]–[27], as well as several tutorial/review papers [28]–[30].

From the point of view of this paper, the arbitrary differentiation order can be interpreted as an alternative means of adjusting the impedance and dissipation characteristics of the overall haptic system. To improve the stability robustness of haptic devices, the conventional approach is to introduce viscous damping, that is proportional to the first derivative of the position signal. However, with a fractional order control approach, the dissipation can be supplied to the system with an arbitrary differentiation order. The order of differentiation affects the dissipation as well as other mechanical impedance characteristics of the fractional order mechanical element. In particular, a differentiation order less than 1 results in a dissipative mechanical element that can store potential energy, while a differentiation order greater than 1 is a dissipative element that can also store kinetic energy [31].

In the literature, many definitions of fractional order differointegrals exists. The most frequently used definitions are the Grunwald-Letnikov, Riemann-Liouville and Caputo's

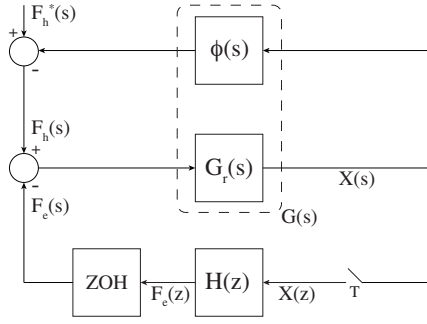


Fig. 1. The sampled-data haptic system with ideal sampler and zero order hold.

definitions. In this paper, Caputo's definition, is preferred due to its advantages in terms of computing the Laplace transformation of functions with fractional order differointegrals.

$${}_a\mathcal{D}_t^\mu f(t) = \frac{1}{\Gamma(n-\mu)} \int_a^t \frac{f^{(n)}(\tau)}{(t-\tau)^{\mu-n+1}} d\tau$$

for $n-1 < \mu < n$.

Using Caputo's definition for fractional order differointegration yields to an important property which vastly reduce the computational burden of the analysis. The Laplace transform of a fractional order differointegral is defined as

$$\mathcal{L}({}_a\mathcal{D}_t^\mu f(t)) = s^\mu \mathcal{L}(f(t))$$

There exist two main approaches for discretization of fractional order differointegration operation. The first approach is direct discretization, where the exact mathematical model of the fractional order differointegral is used for further analysis. These direct discretization methods generally consider series expansions, such as MacLaurin series expansion, power series expansion, and continued fraction expansion. In [32], direct discretization is analyzed and polynomial approximations for arbitrary order differointegration is introduced.

For the indirect method, a mathematical model is fitted to the frequency domain response of the fractional order differointegral. Details of different discretization schemes can be found in [33].

For further details of fractional order calculus, the reader is referred to [25], [34], [35].

IV. HAPTIC SYSTEM

The notation and the structure of the analysis in the rest of the paper closely follows [36], which studies the integer order case. Fig. 1 presents the block diagram of the haptic system in a sampled-data form. The human is represented with the model $\phi(s)$, which may be nonlinear, and an exogenous force $F_h^*(s)$. $G_r(s)$ denotes the haptic display with m_r and b_r denoting the mass and physical viscous damping of the robot. The feedback signal is chosen as the position measurements from the robot and it is sampled with a time period of T . $H(z)$ represents the model of the virtual wall that is implemented on a digital computer. Finally, the computed reaction force, $F_e(z)$ is passed through a zero-order hold and fed back to the plant.

The haptic interface is modeled as a rigid robot and it is assumed that the human operator firmly grasps the robot; hence, $x_h = x_r$. The equations of motion for the robot can be given as

$$m_r \ddot{x}_h + b_r \dot{x}_h = f_h + f_e \quad (1)$$

The corresponding transfer function of the haptic display from force to velocity in continuous time is

$$G_r(s) = \frac{1}{m_r s^2 + b_r s} \quad (2)$$

For coupled stability analysis, the model of the human operator is necessary; however, an accurate model for human operator is generally non-linear, time varying and requires tedious experimentation. On the other hand, a simple LTI mass-spring-damper model is known to be sufficient for studying the some of the main aspects of the presence of a human in the control loop. Therefore, the following model is used for human operator.

$$\phi(s) = m_h s^2 + b_h s + k_h \quad (3)$$

If the sampled-data architecture is manipulated for combining the human and robot models, the following resulting transfer function can be obtained.

$$G(s) = \frac{1}{\underbrace{(m_h + m_r)}_m s^2 + \underbrace{(b_h + b_r)}_b s + \underbrace{k_h}_k} \quad (4)$$

As presented in Fig. 1, in a typical haptic system, the controller is implemented in discrete time and connected to the physical (continuous time) robot via sample and hold mechanisms. In order to perform stability analysis, one has to convert the continuous time systems to its zero-order-hold equivalent discrete time counterpart. Zero-order-hold (ZOH) equivalent representation is preferred, since the real-time implementation of a haptic system is usually carried out using ZOH circuits. Moreover, since the aim of this analysis is to investigate the effect of a fractional order elements in the virtual environment, a simple virtual environment model consisting of a spring and a fractional order linear element is considered as

$$H(z) = K + B \left(\frac{1 - z^{-1}}{T} \right)^\mu \quad (5)$$

In this virtual environment model, K and B are respectively the virtual stiffness and the fractional order linear element parameters of the virtual wall to be rendered. Throughout the analysis, the velocity of the robot is approximated using the backward difference method. Although different velocity approximation methods can be employed for discrete time implementations of the virtual wall, the finite difference approach is chosen due to its simplicity. Moreover, since this approach has been extensively used in the literature, this choice enables a comparison of the performance for integer

and fractional order models. It is important to note that in this virtual wall model, the order of the differentiator is not necessarily an integer number. In this paper, we conduct the analysis for $\mu \in [0, 2]$, where $\mu = 1$ corresponds to the classical backward difference differentiator. We also consider only the positive values of K and B .

Moreover, nondimensional parameters are adapted for the analysis, since as the differentiation order changes, the physical meaning, as well as the unit of B changes. Nondimensional parameters enable comparison of virtual environment models with different differential orders. The nondimensionalization is achieved through the following transformations. Note that these transformations generalizes the ones noted in [9] with the dissipation element. In particular, we generalize the nondimensionalization of virtual damping to fractional order dissipative elements as follows

$$\begin{aligned} K &\rightarrow \alpha = \frac{KT^2}{m}, \quad B \rightarrow \beta = \frac{BT^{2-\mu}}{m}, \\ b &\rightarrow \delta = \frac{bT}{m}, \quad k \rightarrow \gamma = \frac{kT^2}{m} \end{aligned} \quad (6)$$

The ZOH equivalent discrete time counterpart of continuous time system (4) with nondimensional parameters becomes

$$\begin{aligned} G(z) = & \frac{T^2}{m} \frac{((2-c_2-c_3)c_1 + (c_2-c_3)\delta)z}{2c_1\gamma(z^2 - (c_2+c_3)z + e^{-\delta})} \dots \\ & + \frac{(2e^{-\delta} - c_3 - c_2)c_1 + (c_3 - c_2)\delta}{2c_1\gamma(z^2 - (c_2+c_3)z + e^{-\delta})} \end{aligned} \quad (7)$$

where $c_1 = \sqrt{\delta^2 - 4\gamma}$, $c_2 = e^{-(\delta+c_1)/2}$, $c_3 = e^{-(\delta-c_1)/2}$.

In order to perform a stability analysis on the fractional order transfer function, we adopt one of the existing discretization schemes to represent the fractional order elements in terms of their integer order equivalents. Among many well-performing methods, the direct method of Muir recursion introduced in [32] is adopted for the rest of the paper. In particular, the third order polynomial approximation is chosen so that the degree of the differentiator can be kept low, while the approximation error is relatively low. Using this discretization scheme, the fractional order virtual environment model can be written as

$$H(z) = K + B \frac{-\frac{1}{3}\mu z^{-3} + \frac{1}{3}\mu^2 z^{-2} - \mu z^{-1} + 1}{T^\mu} \quad (8)$$

while under this discretization scheme, the nondimensional form of the virtual environment can be given as

$$H(z) = \frac{m}{T^2} \left(\alpha + \beta \left(-\frac{1}{3}\mu z^{-3} + \frac{1}{3}\mu^2 z^{-2} - \mu z^{-1} + 1 \right) \right) \quad (9)$$

Finally, the overall transfer function of the discrete-time system from the exogenous human force to position output of the robot is given by

$$G_x(z) = \frac{G(z)}{1 + G(z)H(z)} \quad (10)$$

V. STABILITY ANALYSIS AND EXPERIMENTAL VERIFICATION

The stability analysis is conducted using the Routh-Hurwitz method. The characteristic polynomial of the closed-loop haptic system with human operator coupled to the haptic display can be found in the denominator of (10). Using bilinear transformation, $z \leftarrow (1+w)/(1-w)$, the discrete-time transfer function is transformed into the proper form for Routh-Hurwitz test. The stability of this new polynomial, which is in w -plane, implies the stability of the discrete-time characteristic polynomial in z -plane.

Experiments for the uncoupled stability of the haptic system is conducted on a single DoF voice coil actuated haptic display shown in Fig. 2.

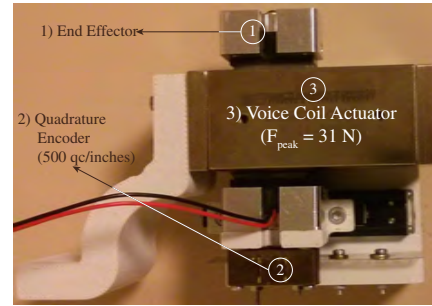


Fig. 2. Single DoF voice coil actuated haptic display

The virtual wall is located at the initial position of the end effector, so that there is no impact during the interaction of the robot with the virtual environment. During the experiments, the robot pushes the virtual wall with a constant force of 1 N and the parameters of the wall are changed until the interaction becomes unstable. The robot interacts with the virtual wall for 5 s and last 0.4 s of this interaction is considered for the analysis. The criteria for determining instability is chosen based on the standard deviation of the end effector position during the last 0.4 s interaction. If the standard deviation is greater than 0.1 mm than the interaction is marked as unstable. This threshold is determined empirically, such that it prevents the end-effector to oscillate significantly, but it is not very conservative allowing for very small amplitude oscillations. In the experiments the sampling time is set as $T = 0.002$ s. The apparent inertia of the robot is $m_r = 65$ g and its physical damping is characterized as $b_r = 3.5 \times 10^{-3}$ N s/mm.

A. Stability Regions

The user imposes on the haptic system an impedance, consisting of stiffness, damping, and mass. Even though this impedance can change based on the user's grip, it necessarily contains relatively low stiffness with high inertia and high damping [10]. One of the worst-case stability scenario takes place when there is minimal damping and low inertia with high stiffness, during which the user is not or is barely touching the haptic device, that is, when the user is adding negligible impedance to the system. A stronger user grip imposes additional damping to the system, augmenting the

natural damping of the device and may help with coupled stability. Hence, in practice, a light/no grip represents one of the most challenging case for stability analysis.

In this section, the stability of the close-loop haptic system is analyzed without involving a human operator, that is, $m_h = 0$, $b_h = 0$ and $k_h = 0$ and only parameters of the robot (m_r and b_r) are utilized. This setup is also important, since it is easier to ensure repeatability of the experimental results. Furthermore, this set up can thoroughly reveal the effect of differentiation order, without complicating the results due to many human hard to model aspects of a human operator.

Stability regions — the regions in the α - β plane, where the closed-loop uncoupled haptic system stays stable — of uncoupled haptic device for various differentiation orders are presented in Fig. 3.

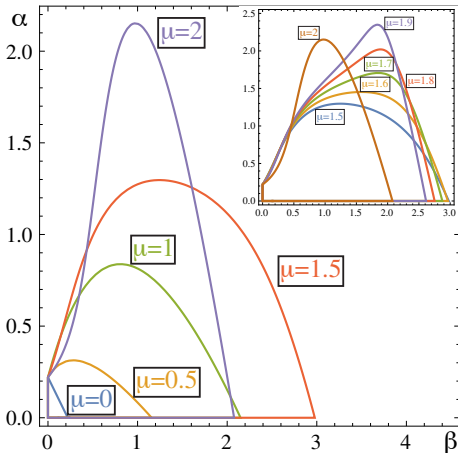


Fig. 3. Stability region of haptic system for various differentiation orders

One can observe from Fig. 3 that, for $\mu \in [0, 2]$ the area under the stability region increases as the differentiation order of the virtual environment increases up $\mu = 2$. It is important to note that when $\mu = 2$, β corresponds to an ideal nondimensional kinetic energy storage (inertia) element and the highest stiffness rendering for the haptic device is achieved when $\mu = 2$. This observation is in agreement with the analysis emphasized in [37] and as expected, the fractional order analysis can perfectly recover the results for the integer order case.

Fig. 4 depicts the results for the Z-width experiments for the voice coil actuated experimental setup introduced in Fig. 2. The experiments are conducted for $\mu \in \{0, 0.5, 1, 1.5, 2\}$, under the same conditions for which the theoretical analysis is conducted.

Comparison of theoretical results presented in Fig. 3 and experimental Z-width results depicted in Fig. 4 reveals that the experimental results indeed capture the effect of differentiation order on the stability region. In particular, as expected from the analysis, the stability region increases as the differentiation order increases. The highest stiffness rendering takes place at the differentiation order $\mu = 2$. The decrease in β during the transition from $\mu = 1.5$ to $\mu = 2$ is also captured by the experiments.

The experimental results successfully capture the qualitative relationship among different differentiation orders as

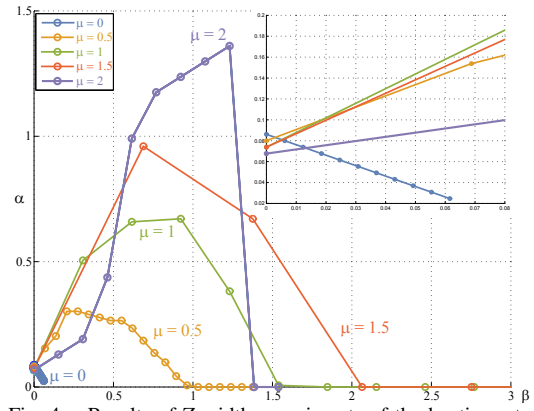


Fig. 4. Results of Z-width experiments of the haptic system

predicted by the analysis. Similarly, the quantitative match is very good for low differentiation orders (less than 10% error for differentiation orders up to 1), but the discrepancy between the theoretical and experimental stability regions increases as the differentiation order increases (displaying a 40% mismatch for $\mu = 2$). The degradation for high differentiation orders is mainly due to quantization noise introduced by the encoder and inherent unstable nature of differentiation operation. In particular, due to the intrinsic noise amplification property of digital derivative estimators, the higher the differentiation order becomes, the higher the quantization noise amplification takes place. A thorough analysis of the noise amplification of fractional order differentiators can be found in [38].

B. Sensitivity of Stability Regions to Changes in Model Parameters

Fig. 5 presents the magnitude plot of the sensitivity function of the fractional order sample-data haptic system for different differentiation orders. One can observe from this figure that as the differentiation order μ gets smaller, the peak magnitude of the sensitivity function decreases, indicating that the system becomes more robust to parameter changes.

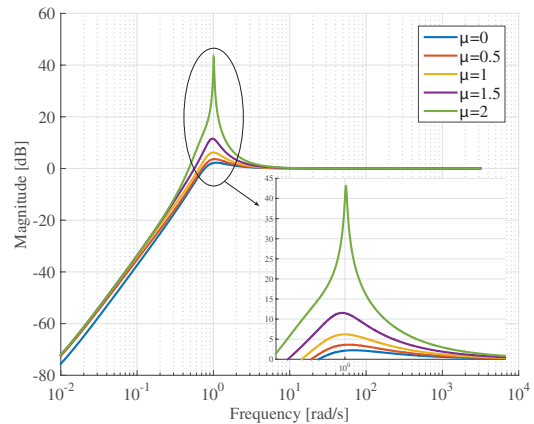


Fig. 5. Sensitivity transfer function of the fractional order haptic system for different differentiation orders μ

Fractional order transfer functions are commonly utilized in robust motion control literature to result in favorable frequency responses [17], since they allow for much larger

range of frequency responses to be synthesized. Fig. 5 is an example of such a case, where decreasing the differentiation order results in favorable stability robustness for the given set of system parameters. In general, even though it is possible to make use of the extra degree of design freedom introduced by the fractional differentiation order into good use to achieve favourable system response, the effect of the fractional differentiation order on the system response is not trivial and strongly depends on system parameters.

VI. CONCLUSIONS

We have proposed utilization of fractional order models/controllers in haptic systems and analyzed the stability of fractional order haptic systems. We have computed stability regions for systems with different orders and experimentally verified these results though a single DoF haptic interface. Our results indicate the non-dimensional stability region enlarges as the differentiation order is increased from 0 to 2. Furthermore, we have observed that fractional order system order can directly affect the stability robustness under parameter variations.

In general, the extra degree of design freedom introduced to the control system by the fractional differentiation order seems promising, since it allows one to better shape the frequency response of the system to achieve more favorable performance characteristics.

Our future works include further study of the effect of fractional order on the sensitivity of the system response to parameter variations and to characterize performance trade-offs that depend on the differentiation order.

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